Indoor Visible Light Positioning Based on Improved Whale Optimization Method With Min-Max Algorithm

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***Abstract*— Recently, the Min-Max algorithm has been widely used in various indoor positioning systems due to its simplicity and robustness. In this article, an improved whale optimization algorithm (IWOA) associated with the Min-Max algorithm is proposed to improve the positioning accuracy concerning the original Min-Max algorithm for visible light positioning (VLP) systems. In the proposed algorithm, the three-dimensional (3-D) region of interest (RoI) affiliated with the target node (TN) is first obtained using the original Min-Max algorithm, and then the IWOA is employed to find the accurate position of the TN in the 3-D RoI with the lowest fitness value. In software simulations, the signal-to-noise ratio (SNR) value of the VLP system is 15 dB and the standard deviation of measured distance noise is 1 m. Our software simulation results show that in the case of two-dimensional (2-D) positioning, the averaged positioning error (APE) of the proposed IWOA-Min-Max algorithm is decreased by 84.6%, 52.1%, 1.9%, and 19.93%, respectively, for the original Min-Max algorithm, the extended Min-Max algorithms, the whale optimization algorithm (WOA), and the least square estimation (LSE). Furthermore, in the case of 3-D positioning, it is also reduced by 81.7%, 75.4%, 8.5%, and 63.9%, respectively, compared with that of the above four existing algorithms. Finally, hardware-in-the-loop simulation (HILS) results are also provided to verify the effectiveness of the proposed IWOA-Min-Max algorithm.**

***Index Terms*— Improved whale optimization algorithm (IWOA), min-max, visible light positioning (VLP).**

I. Introduction

IN RECENT years, the demand for indoor high-precision and real-time positioning has been developed rapidly [1]. The global positioning system (GPS) has the merit of high positioning accuracy in the outdoor positioning scenarios, but it does not perform well in indoor or underground environments [2], [3]. Therefore, various positioning methods, such as Wi-Fi, Bluetooth, radio frequency identification (RFID), ultrawideband (UWB), and visible light positioning (VLP), have been proposed to meet location-based information requirements in indoor environments [4], [5]. Among these methods, the VLP has become a research hotspot due to its wide spectrum, no need to deploy additional infrastructures, insensitivity to electromagnetic interference, widespread usage of green light-emitting diodes (LEDs), the safe and reliable transmission channel, and the excellent positioning performance [2], [6].

The various existing indoor VLP algorithms, such as the least square estimation (LSE) in trilateration [2], fingerprinting, and image-based schemes, have their issues that need to be addressed. For example, the trilateration positioning method has bad performance when the target node (TN) are near the corner [6]. The fingerprint-based method requires the extra collection of the fingerprint data and its positioning accuracy is sensitive to the environmental changes [6], [7]. The image-based positioning methods can only be exploited for the devices equipped with cameras. Recently, Min-Max algorithm has been used in two-dimensional (2-D) wireless sensor networks because of its low-computational complexity and easy implementation [8].

However, the positioning accuracy of the original Min-Max algorithm still needs to be improved, since using this method the location of the TN is only roughly estimated as the geometric centroid of the area of interest determined by the distance between the LED lamp and target. The distance can be converted by received signal strength (RSS) value [8], [9] or time-of-arrival (TOA) parameter [10], [11] measured by the receiver.[1] Therefore, to improve the positioning accuracy of the original Min-Max algorithm, the extended or improved Min-Max algorithms had been recently proposed using the weighted centroid method [12], [13] and area partitioning strategy [14], respectively.

With the development of the bionic optimization technologies, the various bionic intelligent algorithms have been widely used in the VLP systems to improve the positioning performance. For example, a modified artificial fish swarm algorithm was proposed in [15] for the three-dimensional (3-D) VLP system based on the visible light communication (VLC). Moreover, a novel VLC positioning system based on

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[1]In this article, we only consider the RSS-based VLP algorithms. This is because first, the requirement of strict synchronization between the transmitter and the receiver increases the complexity of TOA-based systems; second, the RSS parameters are easy to obtain; and finally, RSS-based systems are low-cost and low-complexity.

the modified particle swarm optimization (PSO) algorithm was presented in [16]. The indoor VLP systems with high positioning accuracy were proposed based on the chaotic PSO algorithm [17] and modified genetic algorithm [18], respectively. In [19], the improved hybrid bat algorithm was proposed to tackle the problems of low positioning accuracy and long latency in the indoor VLP systems. In particular, it is worth mentioning that a whale optimization algorithm (WOA) that mimics the social behavior of humpback whales was proposed in [20]. The WOA only requires lower computation and fewer parameter settings and thus it is widely used in different fields.

Motivated by the intelligent optimization methods mentioned above, the improved WOA (IWOA) is developed to improve the positioning accuracy of the original Min-Max algorithm in this article. The major contributions of this article are summarized as follows.

1) The implementation of the original Min-Max algorithm is extended from 2-D positioning to 3-D scenarios. In the 2-D Min-Max algorithm [8], the horizontal distance between the LED and the TN is used to obtain the rectangular area-of-interest affiliated with the TN. Unlike the 2-D case, in the 3-D-Min-Max algorithm, the slope distance between the LED and the TN needs to be employed to derive the cuboid region-of-interest (RoI) affiliated with the TN. To the best of our knowledge, no similar work about the 3-D-Min-Max algorithm has been studied in the literature.
2) In order to avoid the WOA method falling into local optimum, an IWOA is presented to achieve global optimization. First of all, instead of random initial population distribution used in the WOA method [20], uniform initial population distribution is introduced into the proposed IWOA algorithm so as to achieve a faster convergence speed in early iterations. Moreover, during the subsequent iterations, the nonlinear disturbance is used to ensure the global searching ability of the proposed IWOA algorithm. Finally, during each iteration, a discount factor is randomly assigned to an individual (that is, an estimate of the TN's position) to maintain the diversity of the population.
3) The IWOA based on the Min-Max algorithm, referred to as IWOA-Min-Max in this article, is proposed for indoor VLP systems, in which the 3-D RoI affiliated with the TN is first obtained via the original 3-D-Min-Max algorithm. Then, a uniform initial population is generated in the 3-D RoI using the IWOA method, and the optimal estimate of the TN's position can be obtained by means of iteration.

Notation Statements: For the sake of clarity, throughout this article, scalars are represented by nonbold italics, and vectors are represented by bold letters. The Gaussian distribution with zero-mean and variance σ^2 is shown as $N(0, \sigma^2)$. The uniform distribution between a and b is shown as $\mu(a, b)$. $|\cdot|$ represents the absolute value operator. The $\max(\cdot)$ and $\min(\cdot)$ denote the maximum and minimum functions, respectively.

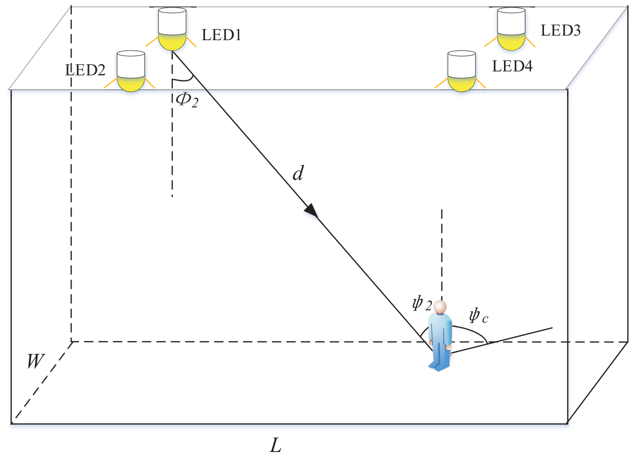


Fig. 1. Room layout of the VLP system.

II. Related Work

A. VLP System Model

A typical room using LED lamps for illumination and positioning is shown in Fig. 1. The length, width, and height of the room are L, W, and H, respectively. All the LED lamps are uniformly installed in the ceiling of the room with the position coordinates $\mathbf{z}_i = [x_i, y_i, H]$, where $i = 1, 2, \ldots, M$ denotes the index of LED lamps. The TN device equipped horizontally with photo detector (PD) is placed in an unknown position $\mathbf{v} = [x, y, z]$. The parameter z changes from 0 to 2 m and it represents the range of average human or robot heights.

In the VLP system, the LED is usually considered as a Lambertian signal propagation model [2]. The LED1 is taken as an example to illustrate the line-of-sight (LOS) propagation model for simplicity. Therefore, the dc gain of the LOS channel and the Lambertian order are given as [2], [6], [21]

$$H(0) = \begin{cases} \dfrac{(m+1)S}{2\pi d^2} G_1(\phi_2, \psi_2), & 0 \le \psi_2 \le \psi_c \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$m = \frac{-\ln 2}{\ln\left(\cos \phi_{\frac{1}{2}}\right)} \quad (2)$$

where $G_1(\phi_2, \psi_2) = \cos^m(\phi_2) T_s(\psi_2) g(\psi_2) \cos(\psi_2)$. $\phi_{1/2}$ is the semi-angle at half illumination of the LED1. S represents the physical area of the PD. d is the distance between the LED1 and the PD. ϕ_2 and ψ_2 denote the angle of irradiance and the angle of incidence with respect to the normal to the LED1's and the PD's surface, respectively. ψ_c represents the field of view angle of the PD. $T_s(\psi)$ is the gain of the optical filter, and $g(\psi)$ is the gain of an optical concentrator, which can be calculated as [6]

$$g(\psi_2) = \frac{n^2}{\sin^2 \psi_c} \quad (3)$$

where n is the refractive index. Assuming the transmitted optical power of each LED is P_t, the electric power received by the PD on the LOS path can be expressed as [2]

$$P_{\mathrm{el}} = R^2 P_t^2 H(0)^2 \quad (4)$$

where R is the responsivity of the PD.

In fact, besides the LOS component of the received signal, the non-line-of-sight (NLOS) components, the thermal, and shot noises can also affect the positioning performance of the RSS-based VLP systems and they can be modeled as Gaussian noise [22]. Thus, the received power can be expressed as

$$P_r = P_{\text{el}} + N_{\text{noise}} \tag{5}$$

where $N_{\text{noise}} \sim N(0, \sigma^2)$.

For the 3-D positioning systems, according to the radiation model in (1), the slope distance between the LED and the TN is given by [6]

$$d_{\text{est}} = \sqrt{\frac{(m+1)S\cos^m(\phi_2)T_s(\psi_2)g(\psi_2)\cos(\psi_2)P_tR}{2\pi\sqrt{P_r}}} \tag{6}$$

$$D_{\text{est}} = d_{\text{est}} + \Delta \tag{7}$$

where D_{est} is the final estimated distance, and $\Delta \sim N(0, \sigma_m^2)$ denotes the measured distance noise [22]. For the 2-D positioning systems, the horizontal distance between the LED and the PD can be calculated as follows [6]:

$$d_h = \sqrt{D_{\text{est}}^2 - (H-z)^2}. \tag{8}$$

B. Original Min-Max Algorithm

In the original Min-Max algorithm, the position of the TN is estimated by calculating the geometric centroid of the vertices in the area of interest [8].

In the 2-D-VLP scenarios, by using the original Min-Max algorithm, the position coordinates of the TN can be approximately estimated as [9]

$$\begin{cases} x = \dfrac{\max\left(x_i - d_h^{(i)}\right) + \min\left(x_i + d_h^{(i)}\right)}{2} \\ y = \dfrac{\max\left(y_i - d_h^{(i)}\right) + \min\left(y_i + d_h^{(i)}\right)}{2} \end{cases} \tag{9}$$

where $d_h^{(i)}$ is the horizontal distance between the TN and the ith LED.

In the 3-D-VLP scenarios, by extending (9) in the 2-D-VLP scenarios, the position coordinates of the TN can be coarsely estimated as follows:

$$\begin{cases} x = \dfrac{x_{lb} + x_{ub}}{2} \\ y = \dfrac{y_{lb} + y_{ub}}{2} \\ z = \dfrac{z_{lb} + z_{ub}}{2} \end{cases} \tag{10}$$

where $x_{lb} = \max(x_i - D_{\text{est}}^{(i)})$, $x_{ub} = \min(x_i + D_{\text{est}}^{(i)})$, $y_{lb} = \max(y_i - D_{\text{est}}^{(i)})$, $y_{ub} = \min(y_i + D_{\text{est}}^{(i)})$, $z_{lb} = \max(H - D_{\text{est}}^{(i)})$, $z_{ub} = z_{\max}$, and $D_{\text{est}}^{(i)}$ is the slope distance between the TN and the ith LED. $z_{\max}$ represents the maximum height of the TN. According to (10), we see that in the 3-D positioning, the ranges of x, y, z are given by $[x_{lb}, x_{ub}]$, $[y_{lb}, y_{ub}]$, and $[z_{lb}, z_{ub}]$, respectively. In other words, using (10) only an approximate estimation of the TN's location can be obtained. Furthermore, we see that a RoI affiliated with the TN can be determined via the original Min-Max algorithm, and the final estimation can be refined by exploiting other methods, such as the weighted centroid method [12], [13] and the area partitioning strategy [14].[2]

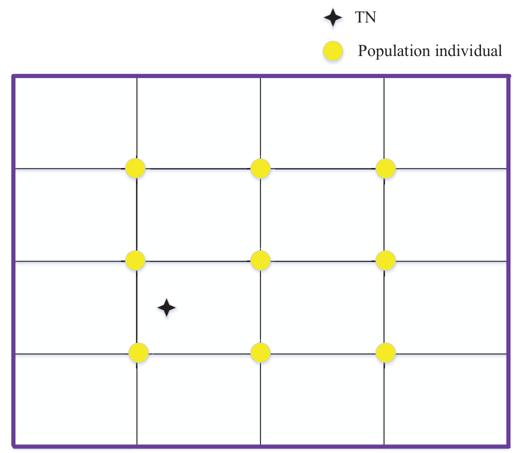


Fig. 2. Initial population distribution in the 2-D positioning systems.

III. Improved WOA

The WOA has been widely used in various fields because of its simple operation and easy implementation [20]. Therefore, in this article, we employ the WOA to improve the positioning accuracy of the original Min-Max algorithm, in which the position of each whale represents the estimated position of the TN. Moreover, the IWOA is presented in this article using the uniform initial population, discount factors, and nonlinear perturbations for achieving the global optimization.

A. Initialization of Population Individuals

For a bionic intelligent algorithm, the population individuals can be usually initialized by the random selection [23] or by the uniform selection [19]. It has been reported that better individuals can be obtained in the early iterations using the uniformly initialized population [19]. Therefore, in this article, the uniform population individuals are considered for the IWOA. Fig. 2 illustrates the 2-D distribution of uniformly initialized population, where the purple rectangle border represents the 2-D boundary of the searching space, which can be obtained by the original Min-Max algorithm. In addition, the initial estimation of the TN's position for the 3-D positioning can be written as $\mathbf{X}_k = (x_1^{(k)}, x_2^{(k)}, x_3^{(k)}), k = 1, 2, \ldots, M_0$, where M_0 denotes the number of the initial populations.

B. Definition of Fitness Function

Assuming that the power vector received at the position $(x_0^{(n)}, y_0^{(n)}, z_0^{(n)})$ by the PD from M_1 LEDs is denoted as $\mathbf{R} = [P_r^{(1)}, P_r^{(2)}, \ldots, P_r^{(M_1)}]$. Due to the limitation of the field of view of the PD, M_1 is less than or equal to M.

[2]Similar to the case of the original Min-Max algorithm, the estimated 3-D coordinates of the TN can also be easily obtained using the extended and improved Min-Max algorithms [12], [13], [14], respectively. Due to the limited space, these results have been omitted in this article.

$\mathbf{D}_{\text{est}} = [D_{\text{est}}^{(1)}, D_{\text{est}}^{(2)}, \ldots, D_{\text{est}}^{(M_1)}]$ can be calculated using (6) and (7).

Therefore, the fitness value at the position $(x_1^{(n)}, x_2^{(n)}, x_3^{(n)})$ in the searching space can be defined as

$$F^{(n)} = \sum_{n=1}^{M_1} \left| C_n - D_{\text{est}}^{(n)} \right| \tag{11}$$

$$C_n = \sqrt{\left(x_1^{(n)} - x_n\right)^2 + \left(x_2^{(n)} - y_n\right)^2 + \left(x_3^{(n)} - H\right)^2}. \tag{12}$$

Algorithm 1 The Proposed IWOA Method

Initialization:

1. Uniformly generate the initial whale population $\mathbf{X}_k$ $(k = 1, 2, \cdots, M_0)$, $t = 0$, and assign values to T_{max}, b, and r.

Procedure:

1: Calculate the fitness of each search agent using (11).
2: Obtain the best position $[X_1^*(0), X_2^*(0), X_3^*(0)]$ based on the minimum fitness criterion.
3: **while** $(t < T_{max})$ **do**
4: Employ discount factor to randomly select an individual whale to update position via (21);
5: **for** $i = 1$ **to** M_0 **do**
6: Update the parameter a using (15), randomly generate r_1, r_2, p_1, p, and calculate A, C;
7: **if** $p < 0.5$ **then**
8: **if** $|A| < 1$ **then**
9: Update individual whale's position using (13) and (14);
10: **if** $p_1 \geq 0.5$ **then**
11: Update whale's position using (18);
12: **end if**
13: **else if** $|A| \geq 1$ **then**
14: Randomly select individual whale to update its position using (19) and (20);
15: **end if**
16: **else if** $p \geq 0.5$ **then**
17: Update individual whale's position using (17);
18: **end if**
19: **end for**
20: Calculate the fitness of each individual, and get the position $[X_1^*(t), X_2^*(t), X_3^*(t)]$;
21: $t = t + 1$;
22: **end while**
23: Return the optical position $[X_1^*, X_2^*, X_3^*]$;
end Procedure

C. Encircling Prey

Humpback whales usually hunt by surrounding their preys first, and this behavior can be modeled as follows [20]:

$$D = \left| C X_i^*(t) - X_i^k(t) \right| \tag{13}$$

$$X_i^k(t+1) = X_i^*(t) - A \cdot D \tag{14}$$

where t represents the current number of iterations. k denotes the kth individual whale in the population. $X_i^k(t)$ and $X_i^k(t+1)$ are the ith position coordinates of the kth individual whale in the tth and $(t+1)$th iteration, respectively, for $i = 1, 2, 3$. $X_i^*(t)$ is the ith position coordinate with the smallest fitness function value in the tth iteration. $A = 2ar_1 - a$, and $C = 2r_2$, where $r_1 \sim \mu(0, 1)$ and $r_2 \sim \mu(0, 1)$. The basis of the WOA for the global and local search depends on A. For a given r_1, A is proportional to a. Thus, to get the global searching ability in the earlier stage of the iteration and the local searching ability in the later stage, a nonlinear transformation method used in the IWOA is adopted

$$a = \frac{2}{1 + e^{\left(t - \frac{T_{\max}}{2}\right)}} \tag{15}$$

where $T_{\max}$ is the maximum number of iterations.

D. Bubble-Net Attacking Method

After surrounding their preys, the humpback whales swim toward to their preys with the spiral motion, and the corresponding mathematical model is as follows [20]:

$$X_i^k(t+1) = X_i^*(t) + D_p e^{b \cdot l} \cos(2\pi l) \tag{16}$$

where $D_p = |X_i^*(t) - X_i^k(t)|$ is the distance between whale and preys. b defines the shape of the spiral and $l \sim \mu(-1, 1)$. At the same time, the whales also need to gradually narrow their encirclement and update the position of whales with the probability selection method as follows [20]:

$$X_i^k(t+1) = \begin{cases} X_i^*(t) - A \cdot D, & p < 0.5 \\ X_i^*(t) + D_p e^{b \cdot l} \cos(2\pi l), & \text{otherwise} \end{cases} \tag{17}$$

where the probability $p \sim \mu(0, 1)$.

The Cauchy mutation in [24], the random walk in [25], and the levy flight strategies in [26] can be adopted to prevent the algorithm from falling into local optimum. If $p_1 \geq 0.5$, the nonlinear perturbation which is similar to the Sinc function is adopted to solve the above problem

$$X_i^k(t+1) = X_i^k(t) + F \tag{18}$$

where p_1 is a random number uniformly distributed between $(0, 1)$. $F = \sin(B)/B$ where $B \sim \mu(-10\pi, 10\pi)$.

E. Searching for Preys

The model of randomly searching for the preys is given as follows [20]:

$$D = \left| C X_i^{\text{rand}}(t) - X_i^k(t) \right| \tag{19}$$

$$D_p = X_i^{\text{rand}}(t) - A \cdot D \tag{20}$$

where $X_i^{\text{rand}}(t)$ represents the ith coordinate of the whale's position randomly selected in the tth iteration. If $|A| \geq 1$, the whales randomly search for preys. Otherwise, the bubble-net method is adopted by whales.

In each iteration of the IWOA, the discounts for previously optimal positions are introduced to randomly update individual whale at tth iteration

$$X_i^R = \frac{\sum_{j=1}^{t-1} r^{t-j} X_i^*(j)}{\sum_{j=1}^{t-1} r^j} \tag{21}$$

TABLE I

SYSTEM SIMULATION PARAMETERS

Symbol	Description	Values
$L \times W \times H$	size of room	5m×5m×3m
R	responsivity of PD	0.53 A/W
$\mathbf{z}$	positions of LEDs [m]	(1.25,1.25,3) (1.25,3.75,3) (3.75,1.25,3) (3.75,3.75,3)
$\phi_{\frac{1}{2}}$	semi-angle at half illuminance	60°
P_t	transmitted optical power of each LED	100 W
n	refractive index	1.5
ψ_c	field of view angle	80°
S	physical area of PD	10 cm^2
T_s	gain of optical filter	1

TABLE II

SIMULATION PARAMETERS OF THE PSO METHOD

Description	Values
particle number	9 ($2D$) 27 ($3D$)
maximum iterations	100
learning factor (c_1)	1.5
learning factor (c_2)	1.5
maximum inertia weight (ω_{max})	0.8
minimum inertia weight (ω_{min})	0.2
maximum flight speed (v_{max})	1
minimum flight speed (v_{min})	-1

TABLE III

PARAMETERS OF THE WOA AND THE PROPOSED IWOA METHODS

Symbol	Description	Values
M_0	number of initial individuals	9 ($2D$) 27 ($3D$)
T_{max}	maximum number of iterations	100
b	shape of the spiral	1
r	discount factor[3]	0.5

where r is the discount factor. X_i^R is the ith coordinate of the position vector of the population in which the serial number is randomly selected.

Based on the above discussions, the implementation process of the proposed IWOA method is presented in Algorithm 1. The proposed IWOA-Min-Max positioning method can be divided into two steps. First, the original Min-Max algorithm is used to estimate the TN's position approximately and form a RoI affiliated with the TN. Then, the proposed IWOA method is employed to find the globally optimal estimation of the TN's position accurately in the RoI.

IV. SOFTWARE SIMULATION RESULTS

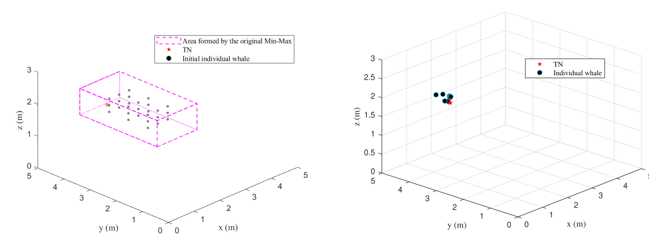
In this section, the performances of the proposed IWOA-Min-Max algorithm and other five candidate algorithms are evaluated using the MATLAB simulation platform. Table I presents the system simulation parameters for all the six investigated algorithms. Among the five counterparts of the proposed IWOA-Min-Max algorithm, the original Min-Max [8], E-Min-Max [12], and LSE [2] algorithms do not require any more parameters except for those listed in Table I. Moreover, the parameters of the PSO [17], WOA [20], and the proposed

[3]The parameter r is exclusively used for the proposed IWOA algorithm.

TABLE IV

DIFFERENT NUMBER OF LED SENSORS SCHEMES[4]

Number of LED sensors	Position (unit: m)
4	(1.25,1.25,3) (1.25,3.75,3) (3.75,1.25,3) (3.75,3.75,3)
5	(1.25,1.25,3) (1.25,3.75,3) (3.75,1.25,3) (3.75,3.75,3) (2.5,2.5,3)
6	(1.6,1.25,3) (1.6,3.75,3) (1.6,2.5,3) (3.4,1.25,3) (3.4,3.75,3) (3.4,2.5,3)
9	(1.25,1.25,3) (1.25,2.5,3) (1.25,3.75,3) (2.5,1.25,3) (2.5,3.75,3) (2.5,2.5,3) (3.75,1.25,3) (3.75,3.75,3) (3.75,2.5,3)
16	(1,1,3) (1,2,3) (1,3,3) (1,4,3) (2;1;3) (2,2,3) (2,3,3) (2,4,3) (3,1,3) (3,2,3) (3,3,3) (3,4,3) (4,1,3) (4,2,3) (4,3,3) (4,4,3)



(a) (b)

Fig. 3. Distributions of individual whales in their initial state and the 100th generation by using the proposed IWOA-min-max algorithm at the sample position (2.0492, 4.3568, 1.4919 m). (a) Initial individual whale. (b) Individual whale in the 100th generation.

IWOA algorithms are shown in Tables II and III, respectively. We randomly select 625 testing samples. The height of the test position is set as 1 m in the 2-D scenarios and it is set a random number uniformly distributed between 0 and 2 m in the 3-D scenarios ($z_{\max} = 2$).

In Figs. 3–5, the coordinates of the TN (2.0494, 4.3568, 1.4919 m) are used as a sample of the TN's position to evaluate the performance of the WOA [20] and the IWOA algorithms.

Fig. 3(a) and (b) show the distributions of individual whales in their initial states and the 100th generations, respectively. The initial population is formed in the area determined by the original Min-Max algorithm and it is evenly distributed. We see that the initial whale population converges around the TN's actual position coordinates after 100 iterations by using the IWOA-Min-Max method.

Fig. 4 presents the convergence of the fitness by using the WOA [20] and the IWOA algorithms with the uniform

[4]For the 2-D positioning, at least two LED sensors are required to obtain a unique estimate for the TN's position in most scenarios [2]. Similarly, in the case of 3-D positioning, it is easy to derive that at least four LED sensors are needed. If the size of the measured room is given, the positioning accuracy will be improved with the increase of the number of LED sensors. However, a trade-off between the positioning accuracy and complexity or cost should be considered in practical system designing. Due to the limited space, the optimization of LED layout designing is not discussed in this article. In the following software and hardware-in-the-loop simulations (HILSs), we mainly considered a measured room with small- or medium-size in which only a minimum number of LED sensors need to be deployed.

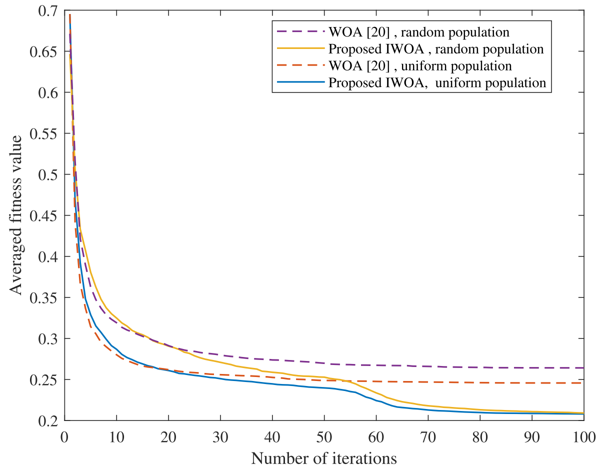


Fig. 4. Convergence of the fitness using the WOA [20] and the proposed IWOA with the uniform population and the random population at the sample position (2.0494, 4.3568, 1.4919 m).

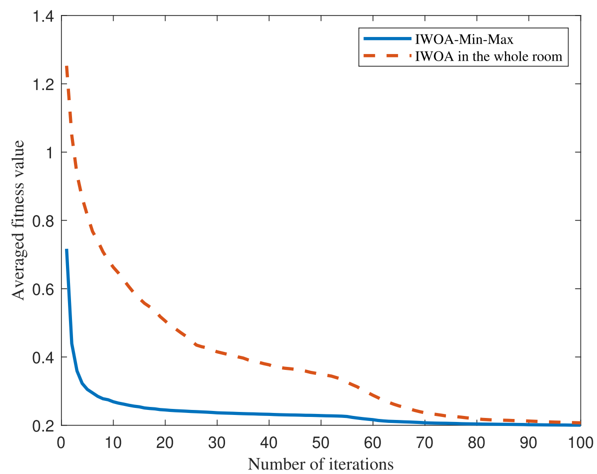


Fig. 5. Convergence of the fitness using the IWOA algorithm in the whole room and the proposed IWOA-min-max method at the sample position (2.0494, 4.3568, 1.4919 m).

population and the random population. For each algorithm, the average results of 100 realizations are used as the final fitness value. With the uniform population, the positioning error of the sample position is equal to 20.84 cm, and 20.35 cm with respect to the WOA [20] and the IWOA, respectively. Meanwhile, with the random population, the positioning error of that is 23.19 and 20.32 cm, respectively. We see that at the beginning of the iteration, the fitness of the uniform population selection decreases faster than that of the random population selection. Moreover, compared with the WOA [20], the proposed IWOA can obtain a lower fitness value at the later stage of iteration, and thus it can achieve better positioning accuracy.

From Fig. 5, it can be seen that the proposed IWOA-Min-Max algorithm quickly finds the globally optimal estimated position of the TN compared with the IWOA used in the whole room under the same initial population number and the system simulation parameters.

Fig. 6 shows the averaged convergence of the fitness using the PSO algorithm [17] and the proposed IWOA-Min-Max method. The averaged fitness value for each iteration can be

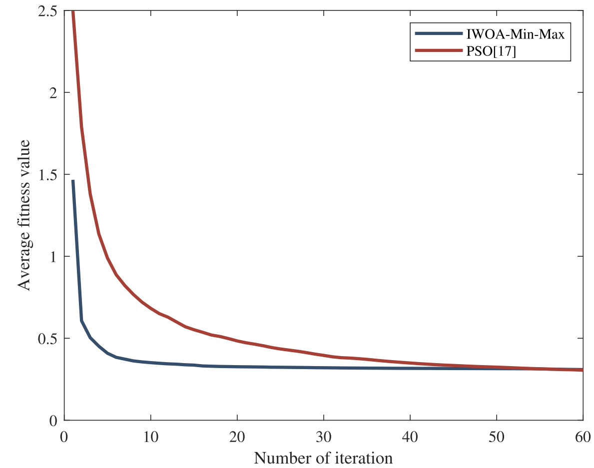


Fig. 6. Averaged convergence of the fitness using the PSO algorithm [17] and the proposed IWOA-min-max method.

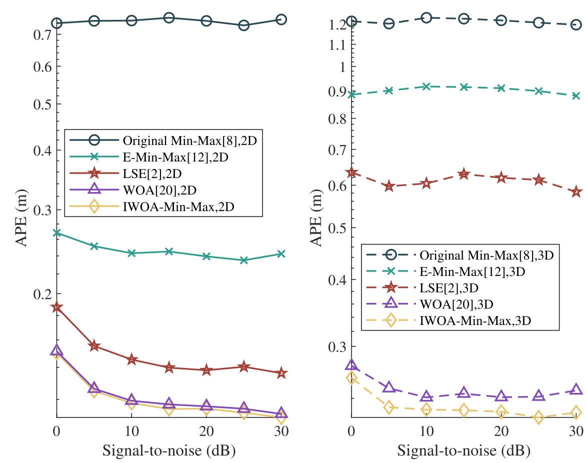


Fig. 7. Impact of SNR for the five investigated algorithms ($\sigma_m = 1$ m).

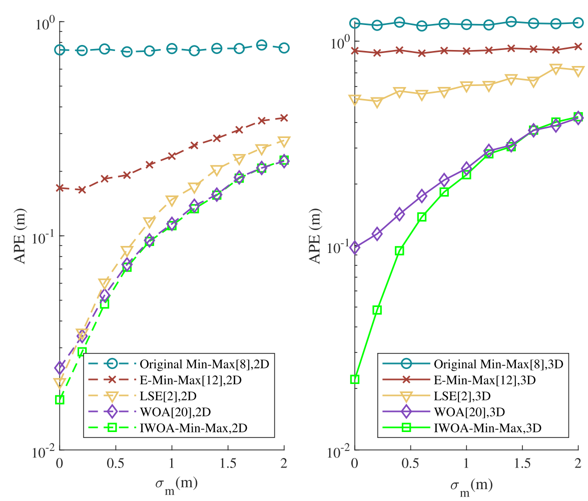


Fig. 8. Impact of σ_m for the five investigated algorithms (SNR = 15 dB).

obtained by averaging the corresponding fitness value at all the test points. We see that for the proposed IWOA-Min-Max, the averaged fitness value can achieve the minimum within 20 iterations. However, for the PSO algorithm, the averaged fitness value can achieve the minimum only after 50 iterations.

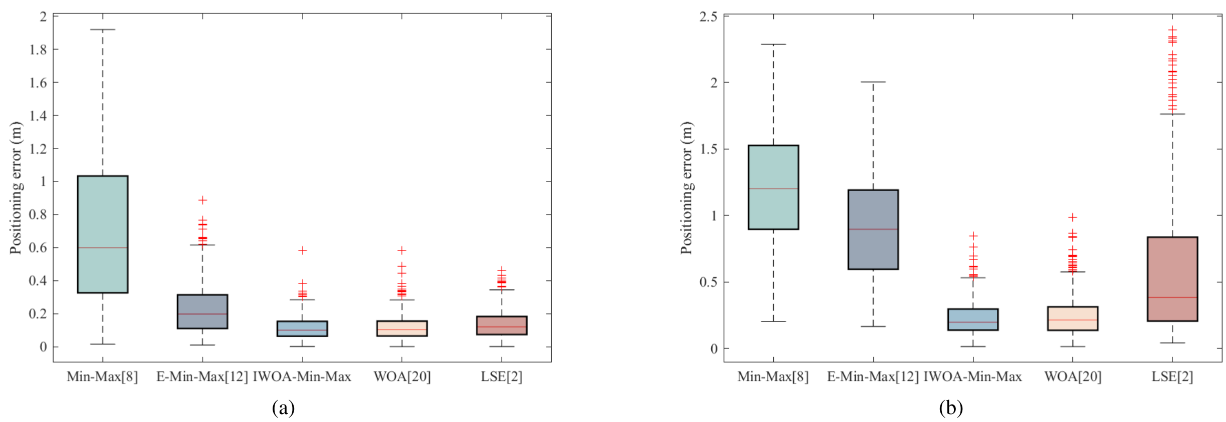


Fig. 9. Boxplot distribution of the five investigated algorithms in (a) 2-D and (b) 3-D positioning when $\sigma_m = 1$ m and SNR = 15 dB.

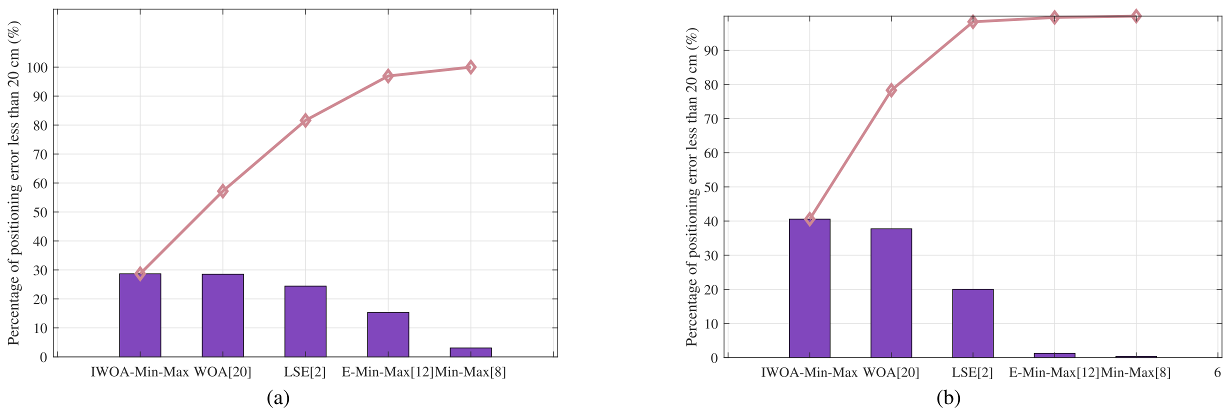


Fig. 10. Pareto charts of the five investigated algorithms in (a) 2-D and (b) 3-D positioning when $\sigma_m = 1$ m and SNR = 15 dB.

Fig. 7 shows the averaged positioning error (APE) of the five investigated algorithms for different signal-to-noise ratio (SNR) values and $\sigma_m = 1$ m. APE refers to the mean of the positioning errors of the 100 implementations at the 625 test positions using the investigated algorithm. We see that when SNR is 15 dB, compared with the original Min-Max [8], the E-Min-Max [12] the WOA [20], and the LSE [2], the APE of the proposed IWOA-Min-Max method is reduced by 84.6%, 52.1%, 1.9%, and 19.93% in the 2-D positioning, respectively. Furthermore, it can be also reduced by 81.7%, 75.4%, 8.5%, and 63.9% in the 3-D positioning.

The APE values of each algorithm increase with the σ_m according to Fig. 8, especially the distance-sensitive E-min-Max algorithm [12], the WOA [20], and the IWOA-Min-Max algorithm. In the 3-D positioning, when the σ_m is set as 0.2 and 2 m, the APE of the IWOA-Min-Max algorithm could reach 4.8 and 42.1 cm, respectively.

Fig. 9 shows the boxplot distribution of the five investigated algorithms in the 2-D and 3-D positioning when $\sigma_m = 1$ m and SNR = 15 dB. We see that the median, third quartile, and maximum positioning error values of the proposed IWOA-Min-Max algorithm are all smaller than those of the other four investigated algorithms in both the 2-D and 3-D positioning cases.

Fig. 10 shows the Pareto charts of the five investigated algorithms in the 2-D and 3-D positioning. For each investigated algorithm, we evaluated the positioning error at 625 test points and counted the frequency with the positioning error of not greater than 20 cm. From Fig. 10, we see that the frequency of the proposed IWOA-Min-Max algorithm accounts for about 30% and 40% in the 2-D and 3-D positioning, respectively, and it is the best among the five investigated algorithms.

In order to evaluate the positioning accuracy of the proposed IWOA-Min-Max algorithm, Table IV presents five schemes with different numbers of LED sensors. Accordingly, Fig. 11 shows the relationship between the number of LED sensors and the APE. We see that as the number of LED sensors increases, the APE of each investigated algorithm generally decreases. We can also see that the proposed IWOA-Min-Max algorithm always has the best positioning accuracy among the five investigated algorithms for different number of LED sensors.

The positioning latency of each investigated algorithm can be calculated by averaging the delay time of the 100 realizations at 625 sample positions, and thus it included time for PD to receive powers from different LEDs and for algorithm positioning. In our evaluation, all five investigated algorithms are performed on a Lenovo laptop configured with an Intel[5] Core[6] I5-5200 CPU at 2.20 GHz. Fig. 12 presents

[5]Registered trademark.
[6]Trademarked.

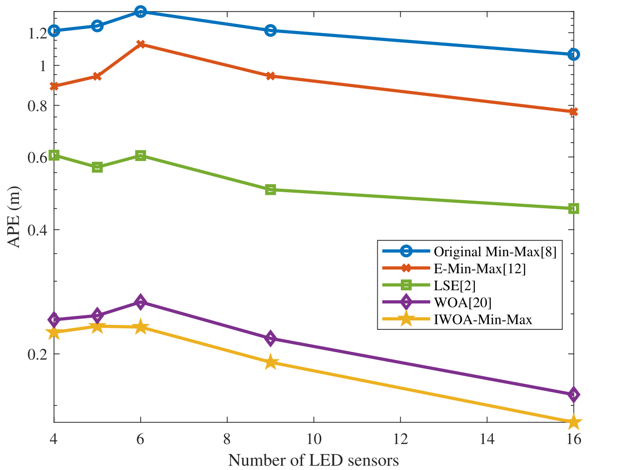


Fig. 11. Relationship between the number of LED sensors and the positioning accuracy when $\sigma_m = 1$ m and SNR = 15 dB.

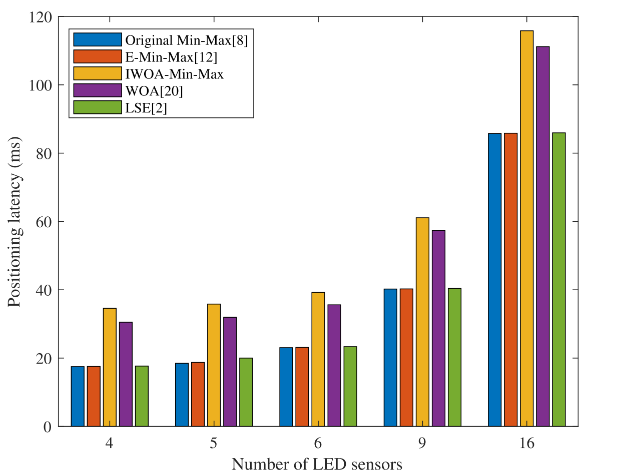


Fig. 12. Relationship between the number of LED sensors and the positioning latency when $\sigma_m = 1$ m and SNR = 15 dB.

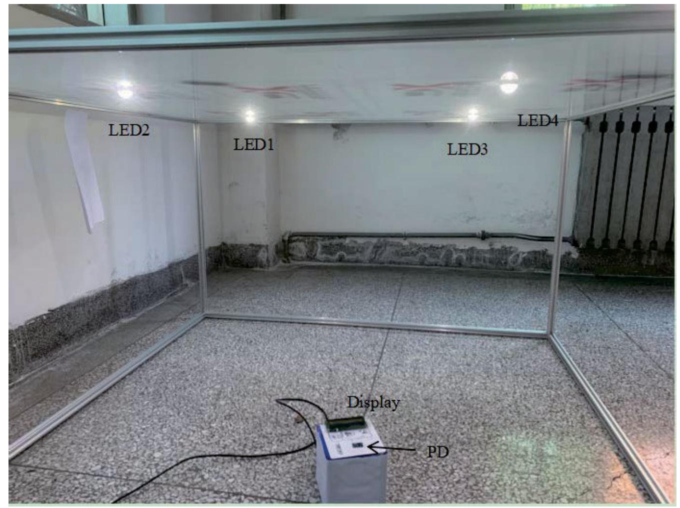


Fig. 13. Layout of the indoor VLP-HILS platform.

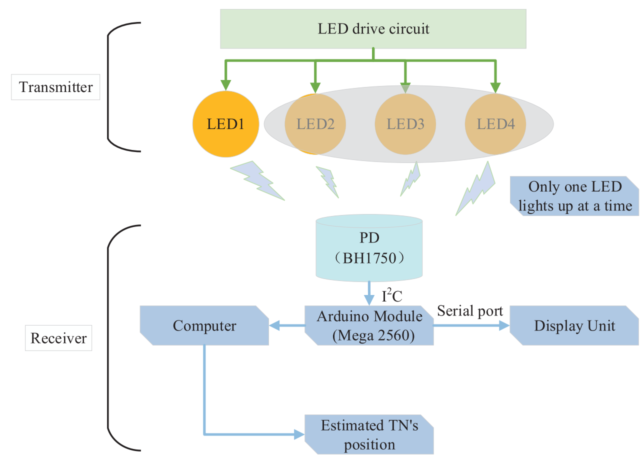


Fig. 14. Block diagram of the indoor VLP-HILS platform.

the positioning latency of the five investigated algorithms with different number of LED sensors. We see that for each investigated algorithm, as the number of LED sensors increases, the positioning latency increases. Moreover, the positioning delays of the original Min-Max [8], the E-Min-Max [12], and the LSE [2] are comparable, and the positioning latency of the proposed IWOA-Min-Max method is the largest. However, when the number of LED sensors is four, the positioning delay of the proposed IWOA-Min-Max method is still less than 40 ms. Therefore, the proposed IWOA-Min-Max method can be suitably acceptable for most real-time applications with small- or medium-size room.

V. HILS Results

In this section, some HILS results are presented to verify the effectiveness of the proposed IWOA-Min-Max algorithm. As shown in Fig. 13, an indoor VLP-HILS is designed to evaluate the positioning accuracy of the proposed IWOA-Min-Max algorithm. Compared with the software simulation, the layout of the VLP-HILS platform is scaled down to $125 \times 125 \times 75$ cm. Four 3-W LED lamps (Model CW332) are evenly installed on the simulated ceiling at the following

TABLE V
Polynomial Fitting Coefficient

	Second order	First order	Constant
LED1	0.0157	-2.1667	89.5555
LED2	0.0627	-4.2353	111.0492
LED3	0.0107	-2.1515	89.8983
LED4	0.0321	-3.2578	102.8571

locations: (31.5, 31.5, 75 cm), (31.5, 94.5, 75 cm), (94.5, 31.5, 75 cm), and (94.5, 94.5, 75 cm). The height of the receiving device (TN) is 10 cm.

The block diagram of the indoor VLP-HILS platform is shown in Fig. 14, where each LED lamp lights up in turn for an intensity test cycle. During each intensity test cycle, the PD first detects the light emitted by the corresponding LED lamp and sends the received light intensity values to the Arduino module. The Arduino module reads the light intensity data and transmits them to the computer for further processing (MATLAB numerical simulation), as well as to the display unit for showing the light intensity simultaneously.

The whole positioning process is divided into two stages: offline fitting and online positioning as shown in Fig. 15. In the

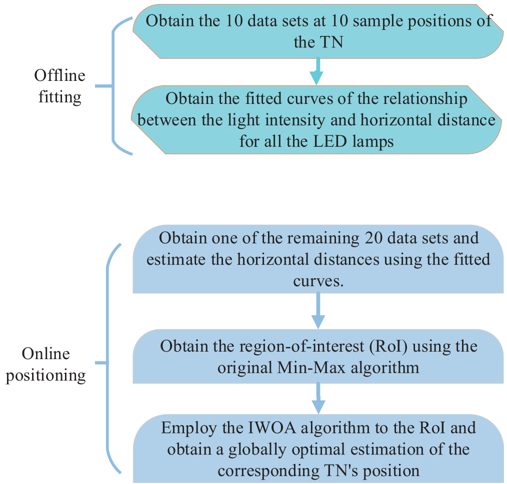


Fig. 15. Flowchart of the indoor VLP-HILS platform.

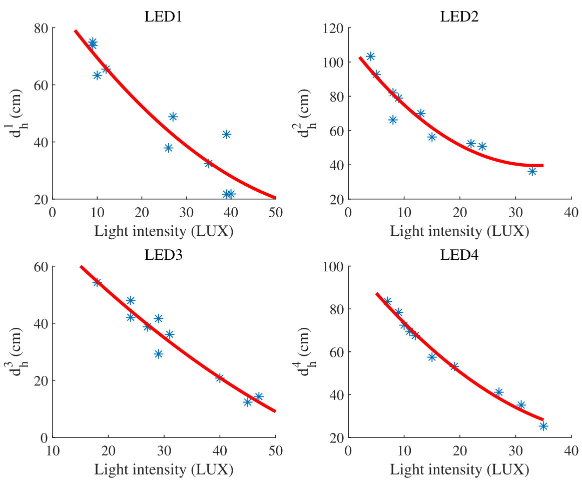


Fig. 16. Fit curves of the relationship between the received light intensity from the ith lamp and the horizontal distance d_h^i, for $i = 1, 2, 3, 4$.

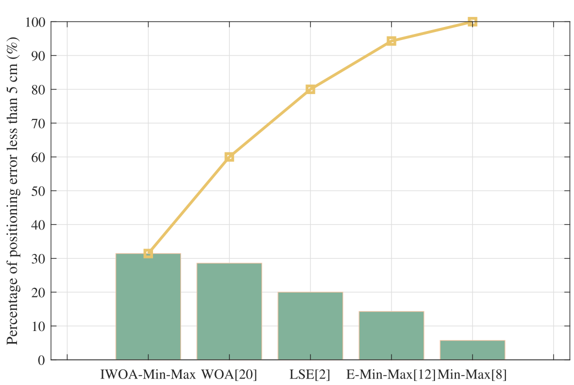


Fig. 17. Pareto chart of the five investigated algorithms (HILS results).

measured area, 30 sets of data are collected at 30 different sample positions of the TN. Each dataset contains the intensity values received from all the four LED lamps as well as the actual position of the TN. Among these datasets, ten of them are used to fit the relationship between the light intensity value received from the ith LED and the horizontal distance d_h^i during the offline fitting stage. The highest degree of the polynomial used for fitting is 2. Fig. 16 and Table V show the fit curves and the corresponding polynomial coefficients, respectively. The remaining 20 datasets are used for testing in the online positioning stage.

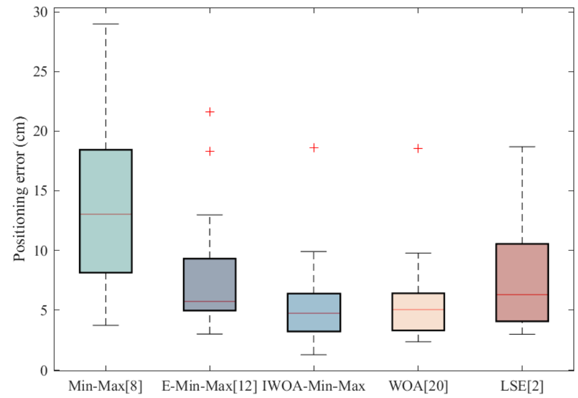


Fig. 18. Boxplot distribution of positioning error of the five investigated algorithms (HILS results).

Fig. 17 shows the Pareto chart via the HILS results for the five investigated algorithms. In the 20 sets of test data, we counted the frequency with the positioning error of less than 5 cm for each investigated algorithm. From Fig. 17, we see that similar to the results shown in Fig. 10, the proposed IWOA-Min-Max algorithm still outperforms other four investigated algorithms. Moreover, Fig. 18 shows the minimum, first quartile, median, third quartile, and maximum positioning error values of the five investigated algorithms. These results further verify the effectiveness and the accuracy of the proposed IWOA-Min-Max algorithm for indoor positioning.

VI. Conclusion

In this article, an indoor positioning method named the IWOA-Min-Max algorithm was proposed to improve the positioning accuracy of the original Min-Max algorithm using the IWOA. In the proposed IWOA-Min-Max algorithm, the TN is first roughly estimated to be located in a region formed by the Min-Max algorithm, and then the optimal estimation position of the TN's position is obtained using the proposed IWOA algorithm in the RoI. Both the software and HILS simulation results show that compared with other candidate algorithms, the proposed method not only has relatively high positioning accuracy but also can meet the real-time requirements. Thus, it can be considered a promising solution for indoor VLP systems.

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